

Lecture 3 – Coverage Criteria

AAA705: Software Testing and Quality Assurance

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2026 Fall

- Random Testing (RT)
 - Geometric distribution – $E(X) = 1/p$, $1 - (1 - p)^n$
 - Residual risk – n clean runs only give $p < \ln(1/\delta)/n$
- Feedback-Directed Random Testing (Randoop)
- Adaptive Random Testing (ART)
 - Distance between inputs, and the 50% upper bound
- Fuzz Testing
 - Generation-based (grammars, Csmith) / mutation-based (bytes, ASTs)
 - **Coverage-guided** fuzzing, and directed fuzzing
 - Sanitizers as oracles, and crash triage

Covered *What*, Exactly?

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- It turns that structure into a finite set of **test requirements**, so “how much have we tested?” becomes a **number**.
- And then the honest question: does a bigger number mean **better tests**? The same question comes back for neural networks.

1. Graph Coverage

- Structural Coverage

- Data-Flow Coverage

- Subsumption Relationships

2. Logic Coverage

- Simple Logic Expression Coverage

- Active Clause Coverage

- Inactive Clause Coverage

- Subsumption Relationships

3. Measuring Coverage

4. Is Coverage a Good Criterion?

5. Coverage Beyond Code

- Neuron Coverage

- Feature-Sensitive Coverage

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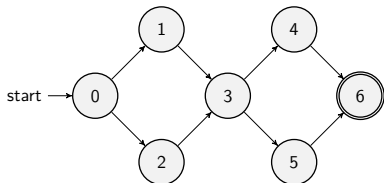
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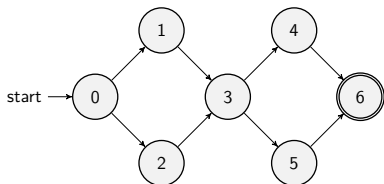
- Neuron Coverage

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int f(int a, int b) {  
    int x = 42;           // 0  
    if (a > 0) g();        // 1  
    else h();              // 2  
    int y;  
    if (b > 0)              // 3  
        y = x * 2;         // 4  
    else                    // 5  
        y = x - 5;         // 5  
    return y;              // 6  
}
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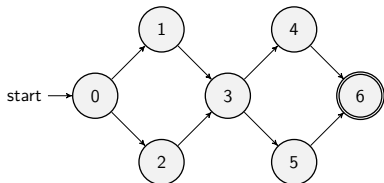


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- A **control flow graph (CFG)** is one of many: **call graphs**, **finite state machines**, class hierarchies, spec structures.
- Define coverage **once** on graphs, and it applies to all of them.

Definition (Graph)

A **graph** $G = (N, E, N_s, N_f)$ consists of a set of **nodes** N , a set of **edges** $E \subseteq N \times N$, a set of **start nodes** $N_s \subseteq N$, and a set of **final nodes** $N_f \subseteq N$.

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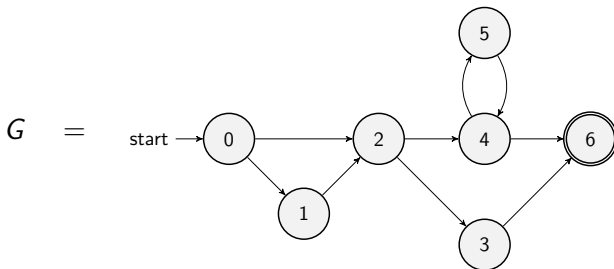
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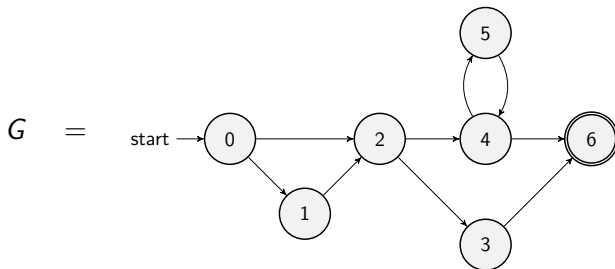
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- A **test path** starts at a start node and ends at a final node
 $(n_0 \in N_s \wedge n_k \in N_f)$ – it is what an **execution** of a test case looks like. We write $\text{path}(t)$ for the test path of a test case t , and $\text{path}(T)$ for those of a test suite T .

Consider a path $p = [0, 2, 4, 5, 4, 6]$ in the graph G .

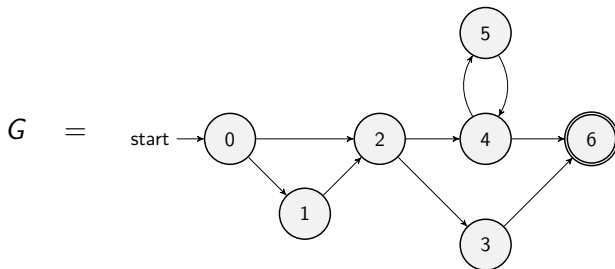


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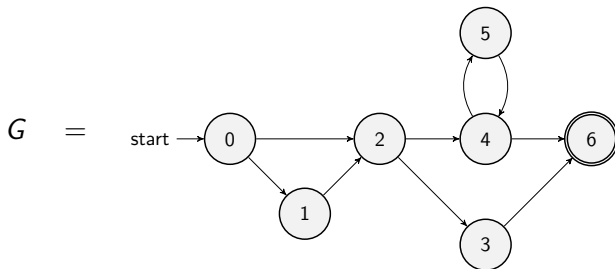
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- $[2, 4, 5] \preceq p$, but $[2, 4, 6] \not\preceq p$ – **contiguous**, not just in order
- p **visits** nodes 0, 2, 4, 5, 6 and edges $(0, 2)$, $(2, 4)$, $(4, 5)$, $(5, 4)$, $(4, 6)$

Definition (Graph Coverage Criterion)

A **graph coverage criterion** $C_G = (R_G, \sim)$ for a graph G is

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- Everything below is a choice of R_G and $\sim$: **structural** criteria use only nodes, edges, and paths; **data-flow** criteria use a graph annotated with **variables**.

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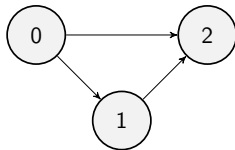
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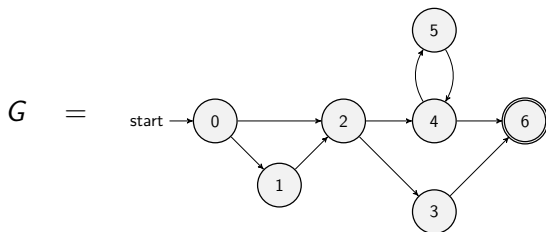
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NC and EC differ only when there is **both** an edge and another subpath between a pair of nodes – an **if** without an **else** is the smallest case: $[0, 2]$ visits every node but misses $(0, 1)$.

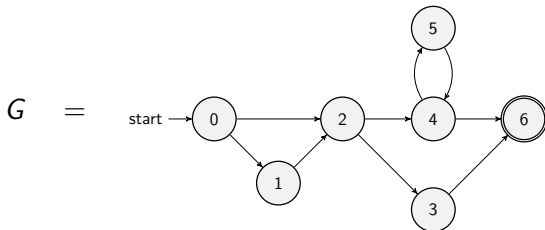




- **Node Coverage (NC)**

$$R_G = \{0, 1, 2, 3, 4, 5, 6\}$$

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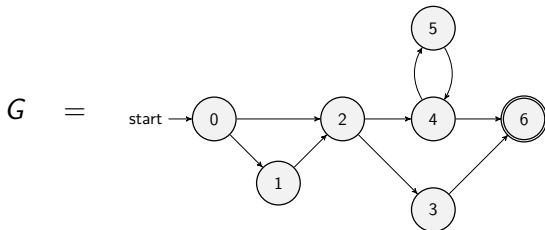
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- The NC suite is **not** EC-adequate: it never takes (0, 2).

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$k = 0$: paths of length 0 = **nodes**
 $k = 1$: + paths of length 1 = **edges**
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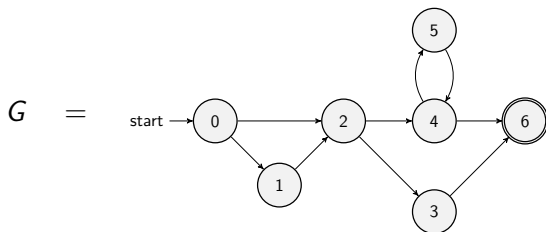
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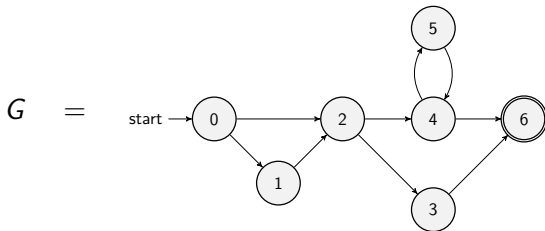
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- Strictly, 1-PC is NC **and** EC together. They coincide as long as every node touches an edge – an **isolated** node is a TR of 1-PC that EC alone would miss.
- So k is a single **knob** trading cost for strength.

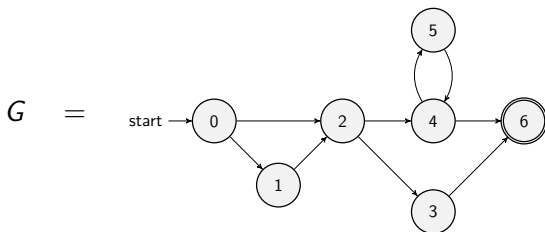
Edge-Pair Coverage – Example





- **11** requirements of length 2 (plus the nodes and edges):

$[0, 1, 2]$, $[0, 2, 3]$, $[0, 2, 4]$, $[1, 2, 3]$, $[1, 2, 4]$, $[2, 3, 6]$,
 $[2, 4, 5]$, $[2, 4, 6]$, $[4, 5, 4]$, $[5, 4, 5]$, $[5, 4, 6]$

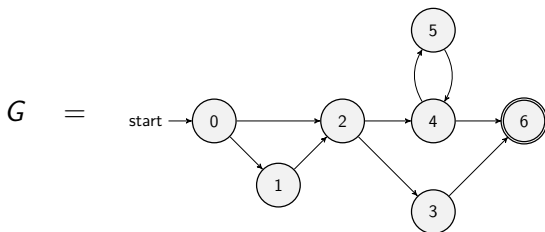


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$= \{[0, 1, 2, 3, 6], [0, 1, 2, 4, 6], [0, 2, 3, 6], [0, 2, 4, 5, 4, 5, 4, 6]\}$



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- Note $[5, 4, 5]$: covering it needs $4 \rightarrow 5 \rightarrow 4 \rightarrow 5$, so EPC forces the loop body **twice**.

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- Four decades of answers:
 - 1970s: execute each cycle once ($[4, 5, 4]$ above)
 - 1980s: execute each loop exactly once
 - 1990s: execute each loop 0 times, once, more than once
 - 2000s: **prime paths** – get all three for free

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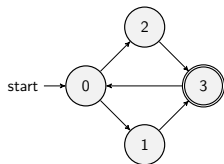
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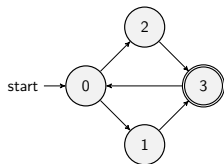
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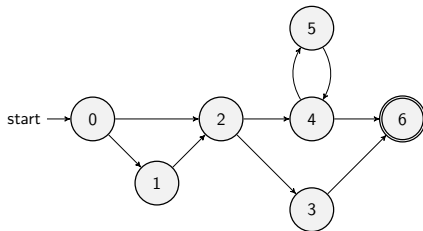
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The other 15 are subpaths of these: [0], [0, 1], [0, 1, 3], ...



38 simple paths, **9** prime:

[0, 1, 2, 3, 6]

[0, 1, 2, 4, 5]

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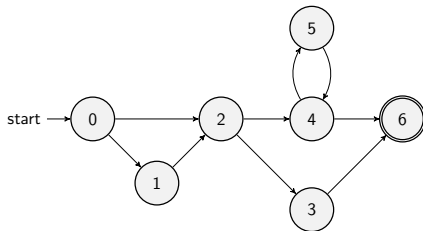
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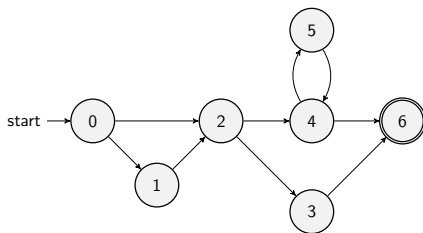
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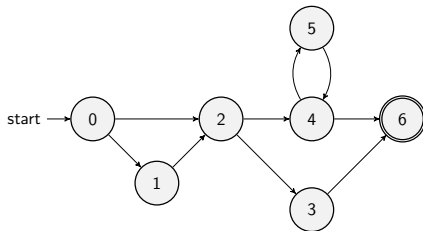
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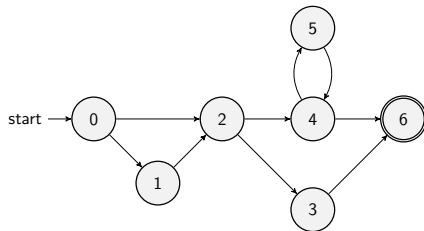
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Prime path coverage needs all three, so it handles loops without a special rule.

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¹[Binder'99] R. V. Binder. "Testing Object-Oriented Systems: Models, Patterns, and Tools." [ISSRE'12] N. E. Holt, R. Torkar, L. Briand, K. Hansen. "State-Based Testing."

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Not every graph is a CFG. In a **state machine** a cycle is checkable: after login $\rightarrow$ logout, **no part of the session may be left behind**.¹

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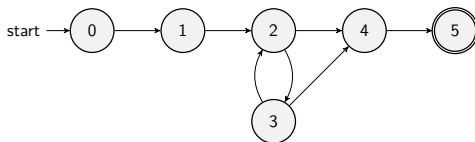
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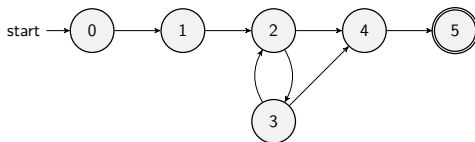
- **Complete round trip coverage (CRTC)** requires **every** round-trip path; **simple round trip coverage (SRTC)** requires **one** per node that has any.
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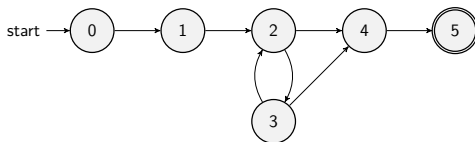
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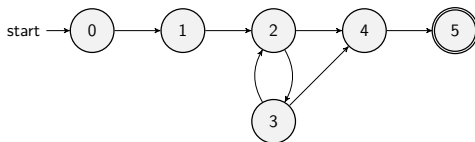
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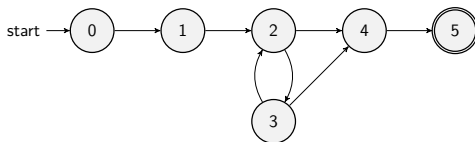
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- A detour may **skip the edge under test** – (2, 4) is never taken above. That is why sidetrips are the usual compromise.

- A test requirement is **infeasible** if **no input** satisfies it: dead code, or a subpath that needs contradictory conditions (`if (x > 0)` inside `if (x < 0)`).

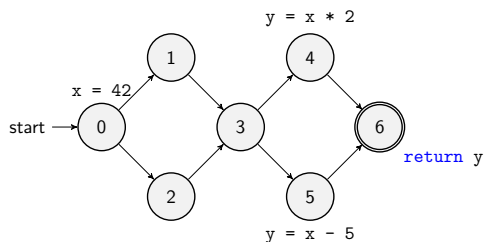
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- **In practice**: satisfy what you can **without** sidetrips, then **allow** sidetrips for the rest, then inspect the remainder by hand.

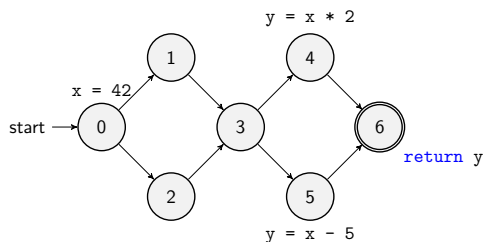
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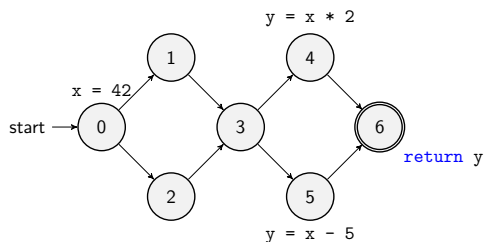
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- Node coverage reaches node 4. It never checks that the 42 written at node 0 is the value node 4 reads.

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	test requirements R_G	p covers it by du-touring
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AUC (<i>all-uses</i>)	(n, n', x)	some $q \in du(n, n', x)$
ADUPC (<i>all-du-paths</i>)	every du-path q	q itself

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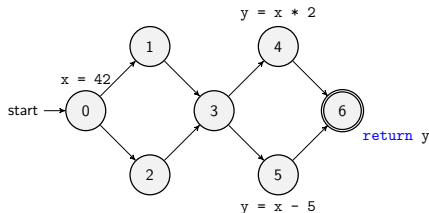
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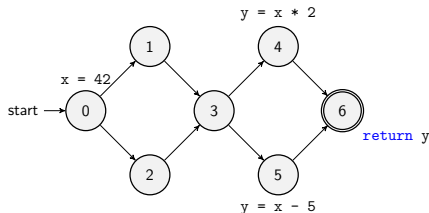
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- So ADUPC is the **strongest** of the three and ADC the **weakest**.
When a tool says “data-flow coverage” it almost always means **AUC** – the other two are too weak and too expensive respectively.



- **All-Defs Coverage (ADC)** – one du-path out of each of the three defs

$$R_G = \{(0, x), (4, y), (5, y)\} \quad \text{Test Paths} = \{[0, 1, 3, 4, 6], [0, 1, 3, 5, 6]\}$$

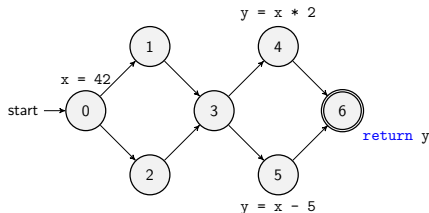


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- **All-DU-Paths Coverage (ADUPC)** – both routes through nodes 1 and 2

$$R_G = \{[0, 1, 3, 4], [0, 1, 3, 5], [0, 2, 3, 4], [0, 2, 3, 5], [4, 6], [5, 6]\}$$
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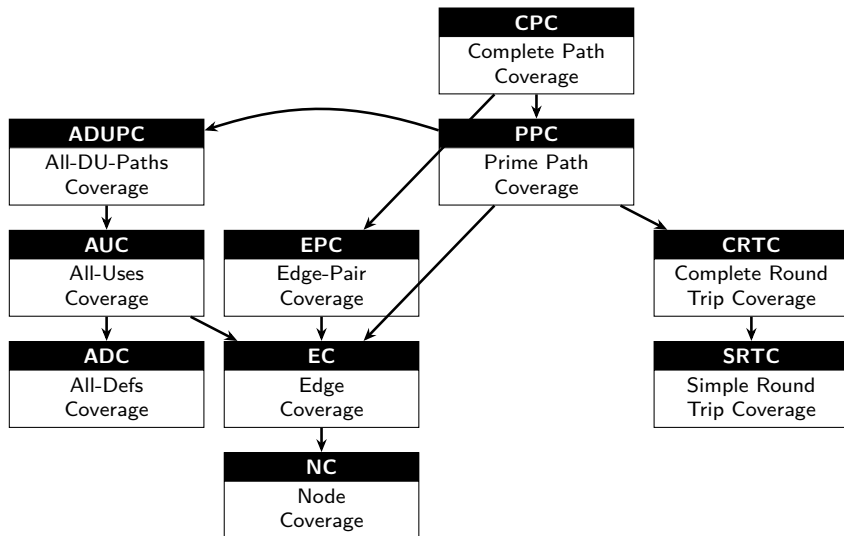
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- Subsumption is about **guarantees**, not about **finding bugs**. A PPC-adequate suite is not automatically better at detecting faults than an EC-adequate one. We come back to this later.



1. Graph Coverage

Structural Coverage

Data-Flow Coverage

Subsumption Relationships

2. Logic Coverage

Simple Logic Expression Coverage

Active Clause Coverage

Inactive Clause Coverage

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3. Measuring Coverage

4. Is Coverage a Good Criterion?

5. Coverage Beyond Code

Neuron Coverage

Feature-Sensitive Coverage

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- Operators: $\neg$, $\wedge$, $\vee$, $\oplus$, $\Rightarrow$, $\Leftrightarrow$.
- Node and edge coverage see **one bit** per **if** – which way it went. Logic coverage looks **inside** the condition.

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For each $p \in P$, the TRs are p evaluating to **true** and to **false**. (Also called **decision coverage**; “branch coverage” in tools.)

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PC asks **two** questions per predicate; CC asks **two per clause**. Both are cheap, and neither implies the other.

$$p = ((a < b) \vee D) \wedge (m \geq n \times o)$$

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PC

a	b	D	m	n	o	p
5	10	T	1	1	1	T
10	5	F	1	1	1	F

CC

a	b	D	m	n	o	$(a < b)$	D	$(m \geq n \times o)$
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- Two tests each. This CC suite **happens** to satisfy PC as well (p is **true** then **false**) – but that is luck, not a guarantee.

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- The simplest fix is to require **every combination**.

Definition (Combinatorial Coverage (CoC))

For each $p \in P$, the TRs are **all combinations** of truth values of the clauses in C_p .

$$p = ((a < b) \vee D) \wedge (m \geq n \times o)$$

$(a < b)$	D	$(m \geq n \times o)$	p
T	T	T	T
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2^N tests for N clauses. We want CoC's **guarantee** at CC's **price**.

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A	B	$A \oplus B$
T	T	F
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A	B	$A \Leftrightarrow B$
T	T	T
T	F	F
F	T	F
F	F	T

- A determines $A \vee B$ only when $B = \text{false}$, and $A \wedge B$ only when $B = \text{true}$.
- A **always** determines $A \oplus B$ and $A \Leftrightarrow B$ – the minor clause cannot mask it.

Definition (Active Clause Coverage (ACC))

For each $p \in P$ and each major clause $c \in C_p$, the TRs are the minor values that make c **determine** p , paired with $c = \text{true}$ and $c = \text{false}$.

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- For $p = A \vee B$:
 - 1 $A = \text{true}, B = \text{false}$ (A determines p)
 - 2 $A = \text{false}, B = \text{false}$ (A determines p)
 - 3 $A = \text{false}, B = \text{true}$ (B determines p)
 - 4 $A = \text{false}, B = \text{false}$ (same row as 2)

$N + 1$ tests instead of 2^N : **three** rows, not four.

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 - 2 $A = \text{false}, B = \text{false}$ (A determines p)
 - 3 $A = \text{false}, B = \text{true}$ (B determines p)
 - 4 ~~$A = \text{false}, B = \text{false}$~~ (same row as 2)

$N + 1$ tests instead of 2^N : **three** rows, not four.

- ACC is better known as **modified condition/decision coverage (MC/DC)**. It is the criterion regulators require for the software that **flies aircraft**.²

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Ambiguity: must the **minor** clauses keep the **same values** when the major clause flips?

$$p = A \vee (B \wedge C), \quad \text{major clause } A$$

A	B	C	p
T	F	T	T
F	F	F	F

is $C = \text{false}$ allowed here?

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Three answers, three criteria:³

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- **General** active clause coverage (GACC)
- **Restricted** active clause coverage (RACC)
- **Correlated** active clause coverage (CACC)

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- GACC does **not subsume predicate coverage**. For $A \Leftrightarrow B$, each clause always determines p , so these two rows satisfy GACC for both A and B – yet p is **true** in both.

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F	F	T

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A	B	$A \Leftrightarrow B$
T	T	T
F	F	T

- GACC was meant to be **stronger** than predicate coverage. It is not even as strong. So we need a different rule.

RACC

A	B	C	$A \wedge (B \vee C)$
T	T	T	T
F	T	T	F
T	T	F	T
F	T	F	F
T	F	T	T
F	F	T	F

CACC

A	B	C	$A \wedge (B \vee C)$
T	T	T	T
T	T	F	T
T	F	T	T
F	T	T	F
F	T	F	F
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F	T	T	F
F	T	F	F
F	F	T	F

- **RACC** (*restricted*) pairs rows: B and C must be identical across the flip, so only **three** pairs are legal.

RACC

A	B	C	$A \wedge (B \vee C)$
<i>T</i>	<i>T</i>	<i>T</i>	<i>T</i>
<i>F</i>	<i>T</i>	<i>T</i>	<i>F</i>
<i>T</i>	<i>T</i>	<i>F</i>	<i>T</i>
<i>F</i>	<i>T</i>	<i>F</i>	<i>F</i>
<i>T</i>	<i>F</i>	<i>T</i>	<i>T</i>
<i>F</i>	<i>F</i>	<i>T</i>	<i>F</i>

CACC

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<i>T</i>	<i>T</i>	<i>F</i>	<i>T</i>
<i>T</i>	<i>F</i>	<i>T</i>	<i>T</i>
<i>F</i>	<i>T</i>	<i>T</i>	<i>F</i>
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<i>T</i>	T	T	T
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CACC

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<i>T</i>	T	T	T
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- **CACC** (*correlated*) only requires p to flip, so **any** top row pairs with **any** bottom row – **nine** choices.
- **RACC** is what the aviation industry adopted. **CACC** is the later reading: weaker, easier to satisfy, and still enough to subsume predicate coverage.

Definition (Inactive Clause Coverage (ICC))

For each $p \in P$ and each major clause $c \in C_p$, the TRs are the minor values that make c **not determine** p , paired with $c = \text{true}$ and $c = \text{false}$.

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A	B	$A \vee B$	$A \oplus B$	
T	F	T	T	ACC
F	F	F	F	ACC
T	T	T	F	ICC
F	T	T	T	ICC

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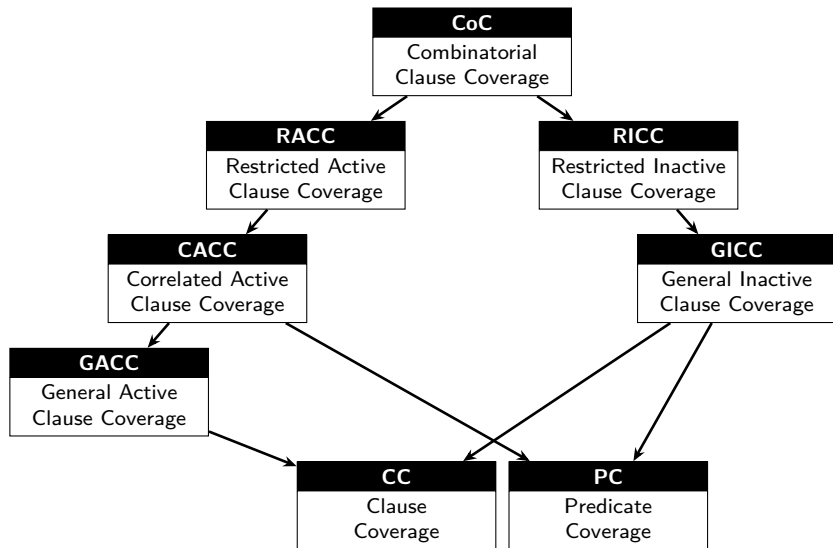
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- Same ambiguity: **GICC** / **RICC**. No “correlated” variant – CACC asks the minor clauses to make p **flip** with c , and under ICC p never flips.



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Neuron Coverage

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$$\text{coverage}(T, C_G) = \frac{|\{r \in R_G \mid \exists t \in T. t \sim r\}|}{|R_G|}$$

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- The denominator is **not stable**: adding a line changes $|R_G|$, so the percentage moves without anyone writing a test.
- Three questions remain: **where** do the numbers come from, **what** do real tools count, and **what does the number mean?**

To know that node 4 ran, the program must **say so**. Insert probes:

```
int f(int a, int b) {  
    cov[0] = 1; int x = 42;  
    if (a > 0) { cov[1] = 1; g(); }  
    else      { cov[2] = 1; h(); }  
    cov[3] = 1; int y;  
    if (b > 0) { cov[4] = 1; y = x * 2; }  
    else      { cov[5] = 1; y = x - 5; }  
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- Coverage asks **whether** a block ran, not how often – so a probe can be **removed** once it fires.⁴

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- “Branch coverage” in a tool means **predicate coverage** on each decision – not edge coverage of the whole CFG, and not MC/DC.
- A fuzzer goes further: AFL **hashes** edges into 64K buckets, so two edges can share one. It wants a **fast** signal, not an exact one – the criterion follows from the **purpose**.

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100% Coverage Is Not Correctness

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TEST(FibTest, Smoke) {  
    fib(10);           // 100% line and branch coverage of fib  
}
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- **Necessary, not sufficient**. So how much does the necessary condition buy?

- Bigger suites cover more **and** find more. Control for suite size and the correlation with fault detection is **weak** – and the studies still argue about how weak.⁵

Low coverage is reliable bad news; high coverage is weak good news.

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- **As a search gradient.** Last week's fuzzer, and techniques still to come. Nobody reports the percentage; it is only a signal.
- **Not as a quality measure.** For that, seed faults and count kills – **mutation score**, which we take up later.

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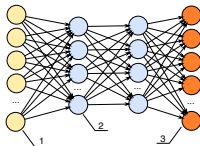
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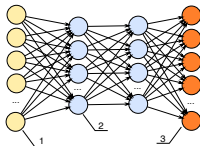
Neuron Coverage

Feature-Sensitive Coverage



- A DNN classifier is a function $f_\theta : X \rightarrow Y$ whose behavior lives in the **weights**, not in control flow. There is nothing to branch on – so cover the **neurons** instead. Write $f_\theta(n, x)$ for the output of neuron n on input x ; the neuron is **activated** when that exceeds a threshold t .

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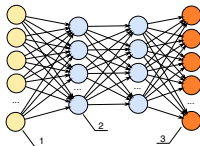


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Definition (Neuron Coverage (NC))

$$NC(T, t) = \frac{|\{n \in N \mid \exists x \in T. f_\theta(n, x) > t\}|}{|N|}$$

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Plain NC **saturates**, so it was refined by splitting each neuron's training range into k sections (**KMNC**), or looking outside that range (**NBC**), or taking the top k per layer (**TKNC**) – **the same k knob again.**⁷

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Does Neuron Coverage Work?

⁸[FSE'20] F. Harel-Canada, L. Wang, M. A. Gulzar, Q. Gu, M. Kim. *"Is Neuron Coverage a Meaningful Measure for Testing Deep Neural Networks?"*

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Unfortunately, no!

- Generate inputs that **maximize** NC and see what you get: **fewer** defects detected, **less natural** inputs, and **more biased** predictions. Higher NC made the suite **worse**.⁸

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- Generate inputs that **maximize** NC and see what you get: **fewer** defects detected, **less natural** inputs, and **more biased** predictions. Higher NC made the suite **worse**.⁸
- Independently, the criteria correlate poorly with model quality.⁹

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- Same story as code coverage: **a criterion has to be validated, not just defined.**

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```
a = [10, 20, 30];  
a.at(-1); // 30  
a.at(5);  // undefined
```

23.1.3.1 **Array.prototype.at** (*index*)

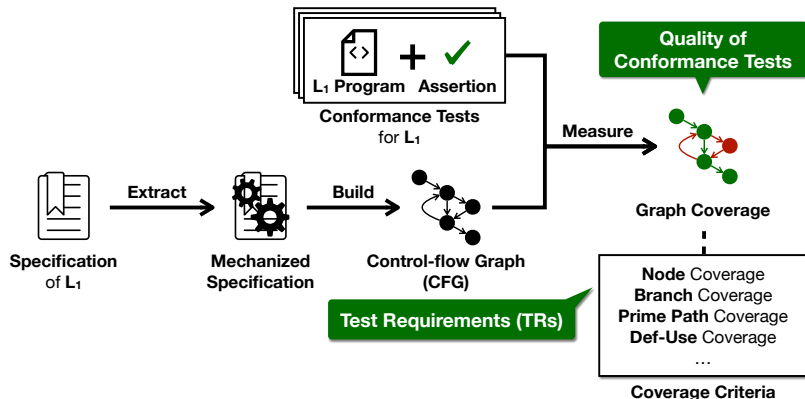
1. Let *O* be ? **ToObject**(*this* value).
2. Let *len* be ? **LengthOfArrayLike**(*O*).
3. Let *relativeIndex* be ? **ToIntegerOrInfinity**(*index*).
4. If *relativeIndex* ≥ 0 , then
 - a. Let *k* be *relativeIndex*.
5. Else,
 - a. Let *k* be *len* + *relativeIndex*.
6. If *k* < 0 or *k* $\geq$ *len*, return **undefined**.
7. Return ? **Get**(*O*, ! **ToString**($\mathbb{F}(k)$)).

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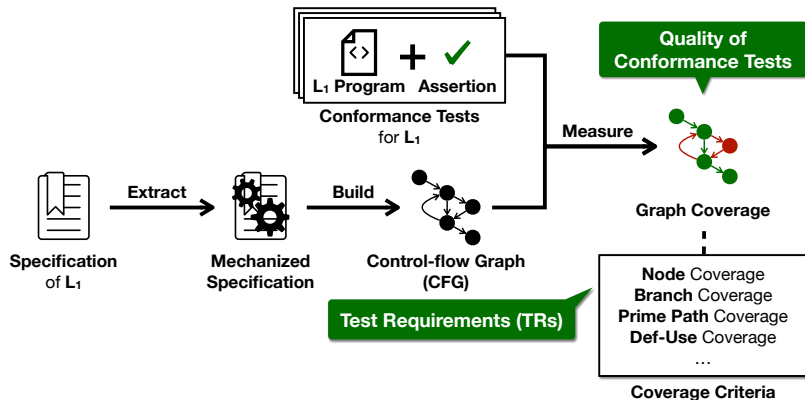
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6. If *k* < 0 or *k* $\geq$ *len*, return **undefined**.
7. Return ? Get(*O*, ! ToString($\mathbb{F}(k)$)).

- ECMA-262 defines JavaScript as **pseudocode algorithms**. The two calls take **different paths** through this one: `at(-1)` runs $5 \rightarrow 7$, `at(5)` runs $4 \rightarrow 6$.

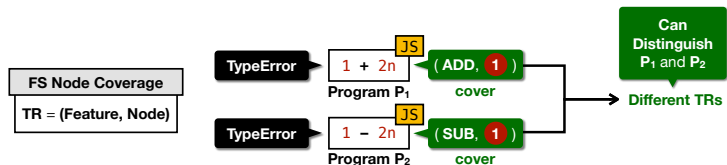


¹¹[ICSE'21] J. Park, S. An, D. Youn, G. Kim, S. Ryu. "JEST: N+1-Version Differential Testing of Both JavaScript Engines and Specification."

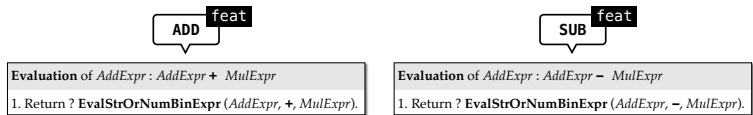


- **JEST**: 1,700 programs, **87.70%** of ES11 semantics – **44 engine bugs, 27 spec bugs**.¹¹

¹¹[\[ICSE'21\]](#) J. Park, S. An, D. Youn, G. Kim, S. Ryu. "JEST: N+1-Version Differential Testing of Both JavaScript Engines and Specification."

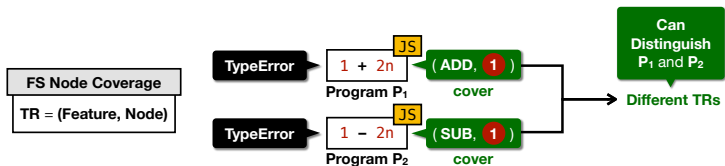


- Feature-Sensitive (FS) coverage criterion divides the given TRs with the innermost enclosing language features

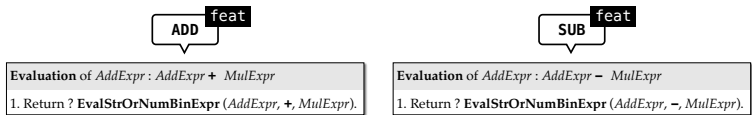


¹²[PLDI'23] J. Park, D. Youn, K. Lee, S. Ryu. "Feature-Sensitive Coverage for Conformance Testing of Programming Language Implementations."

¹³[TOSEM'26] K. Lee, S. Kim, J. Park, S. Ryu. "Selective Feature-Sensitive Coverage for Conformance Testing of Programming Language Implementations."



- Feature-Sensitive (FS)** coverage criterion **divides** the given TRs with the **innermost enclosing** language **features**



- 143** bugs in eight implementations.¹² Selective FS: **70.7%** fewer tests but preserves 53 of 57 bugs.¹³

¹²[PLDI'23] J. Park, D. Youn, K. Lee, S. Ryu. "Feature-Sensitive Coverage for Conformance Testing of Programming Language Implementations."

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1. Graph Coverage

- Structural Coverage

- Data-Flow Coverage

- Subsumption Relationships

2. Logic Coverage

- Simple Logic Expression Coverage

- Active Clause Coverage

- Inactive Clause Coverage

- Subsumption Relationships

3. Measuring Coverage

4. Is Coverage a Good Criterion?

5. Coverage Beyond Code

- Neuron Coverage

- Feature-Sensitive Coverage

- Build a **coverage profiler** that measures which parts of a program are executed by a test suite.
- Please see this document on GitHub:

<https://github.com/ku-plrg-classroom/js-cov>

- The due date is **23:59 on September 21 (Mon.)**, and please submit it to **LMS**.

- Search Based Software Testing

Jihyeok Park

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